

Comparing social needs for transport, gaps in transit service and the Mobility Index for the County Sligo

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Introduction

Research Background

GTFS data

Data for this paper was sourced from Transport for Ireland¹ on October 1, 2025². The analysis was run for the week starting October 1.

¹ “Public Transport Data” (National Transport Authority, undated), https://www.transportforireland.ie/transitData/PT_Data.html.

² Selecting the “All GTFS Operators” feed

Results

Table 1: Summary statistics; SI score by transit supply category, by Small Area.

Characteristic	Overall N = 310	Transit supply category						
		Zero Supply N = 46	Very Low N = 56	Low N = 63	Below average N = 73	Above average N = 38	High N = 26	Very High N = 8
SI								
Min	0.0	0.0	0.0	2.4	41.0	419.0	986.6	2,257.1
Q1	0.6	0.0	0.0	4.9	100.1	536.1	1,210.8	2,387.5
Mean	321.5	0.0	0.6	14.0	201.1	652.4	1,467.7	2,637.9
Q3	374.9	0.0	1.1	24.2	300.9	733.7	1,737.7	2,855.7
Max	3,237.0	0.0	2.2	35.9	408.4	978.1	2,125.3	3,237.0
Sum	99,651.0	0.0	34.9	883.2	14,678.8	24,791.7	38,159.6	21,102.9
population								
Min	74	118	74	97	81	99	141	78
Q1	181	160	175	190	191	199	206	107
Mean	226	206	209	239	240	224	241	212
Q3	261	229	249	272	284	248	278	236
Max	574	362	359	408	450	387	374	574
Sum	70,198	9,461	11,714	15,053	17,502	8,512	6,262	1,694
Urban_area								
Sligo	86 (28%)	0 (0%)	0 (0%)	2 (3.2%)	20 (27%)	30 (79%)	26 (100%)	8 (100%)
Tubbercurry	9 (2.9%)	1 (2.2%)	0 (0%)	4 (6.3%)	4 (5.5%)	0 (0%)	0 (0%)	0 (0%)
Strandhill	7 (2.3%)	0 (0%)	0 (0%)	0 (0%)	5 (6.8%)	2 (5.3%)	0 (0%)	0 (0%)
Collooney	7 (2.3%)	0 (0%)	0 (0%)	1 (1.6%)	5 (6.8%)	1 (2.6%)	0 (0%)	0 (0%)
Ballysadare	9 (2.9%)	0 (0%)	0 (0%)	0 (0%)	6 (8.2%)	3 (7.9%)	0 (0%)	0 (0%)
Ballymote	7 (2.3%)	0 (0%)	1 (1.8%)	0 (0%)	6 (8.2%)	0 (0%)	0 (0%)	0 (0%)
Enniscrone	7 (2.3%)	4 (8.7%)	0 (0%)	2 (3.2%)	1 (1.4%)	0 (0%)	0 (0%)	0 (0%)
Coolaney	5 (1.6%)	2 (4.3%)	0 (0%)	0 (0%)	3 (4.1%)	0 (0%)	0 (0%)	0 (0%)
Rosses Point	4 (1.3%)	0 (0%)	0 (0%)	0 (0%)	2 (2.7%)	2 (5.3%)	0 (0%)	0 (0%)
Grange	2 (0.6%)	0 (0%)	0 (0%)	0 (0%)	2 (2.7%)	0 (0%)	0 (0%)	0 (0%)
Cliffony	3 (1.0%)	0 (0%)	0 (0%)	1 (1.6%)	2 (2.7%)	0 (0%)	0 (0%)	0 (0%)
Other	16 (5.2%)	1 (2.2%)	0 (0%)	2 (3.2%)	13 (18%)	0 (0%)	0 (0%)	0 (0%)
Non-urban	148 (48%)	38 (83%)	55 (98%)	51 (81%)	4 (5.5%)	0 (0%)	0 (0%)	0 (0%)

¹ n (%)

Table 1: Summary statistics; SI score by transit supply category, by Small Area.

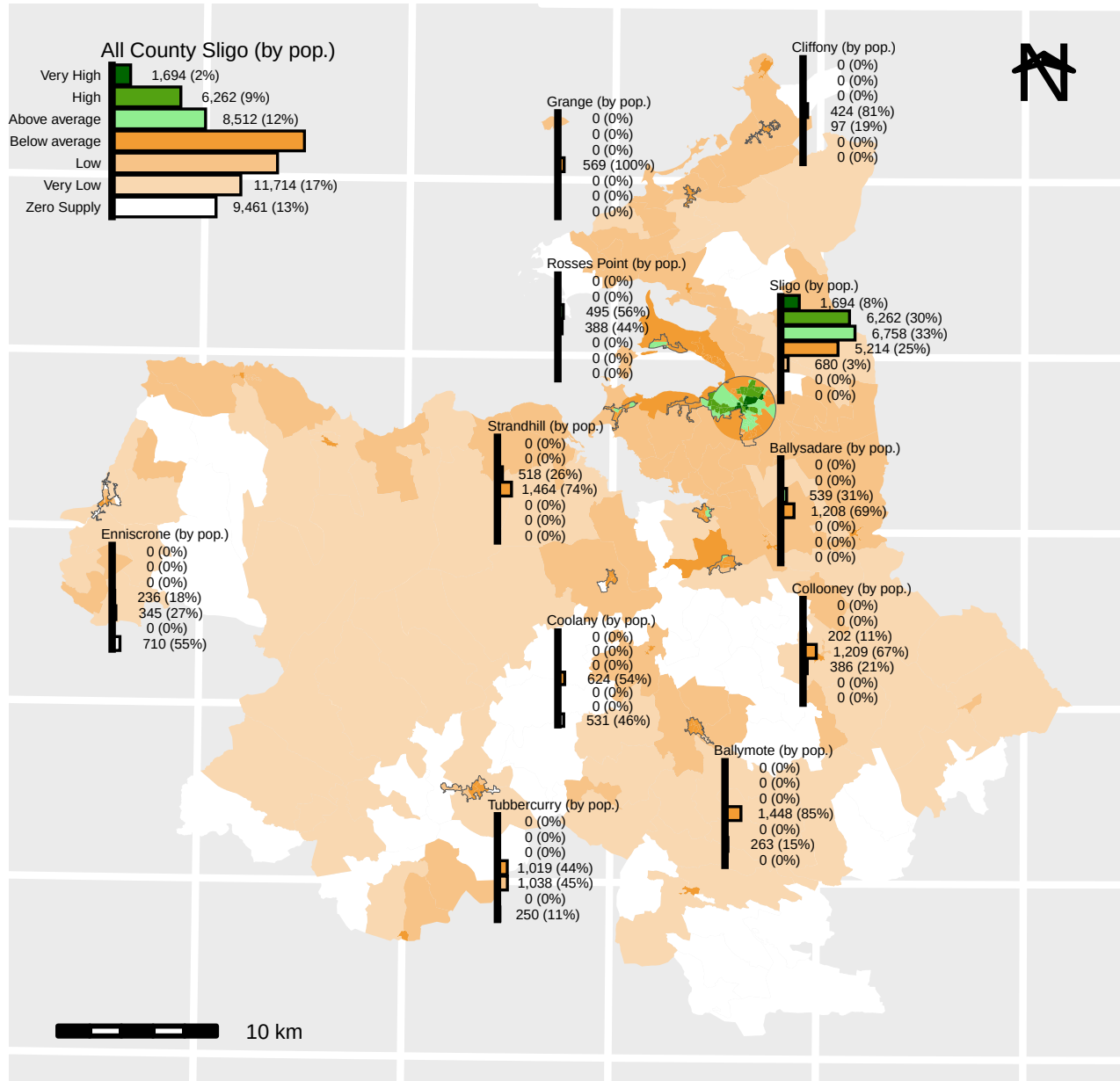


Figure 1: Transit Supply Index categories across County Sligo, by Small Area and by population (for urban areas with > 500 people)

Table X shows summary statistics for SI scores and the population, overall for all of the County Sligo and split by transit supply category. The total population of County Sligo is 70,198, of whom 77% (53,730) live in Small Areas where the SI score is below the average (of those Small Areas with a non-zero SI score) (377.5). This includes the 9,461 people (13%) who live in Small Areas with no transit at all.

Figure 1 shows transit Supply Index (SI) categories by Small Areas across County Sligo. It also shows to the population living in Small Areas in each category for both the entire County and for various urban areas. In general, there appear to be large areas of County Sligo with SI scores below the average, or with no transit supply at all. Also, as might be expected, Small Areas where SI scores are higher than the average (of those with non-zero scores) appears to be located in very small and scattered geographic areas (at population centers).

Table 2 summarises the Small Area populations by Urban Area in County Sligo by transit supply category, shown as a share of the total County population. Table 3 shows SI scores and transit supply categories by Urban Area, weighted by population.

Table 2: County Sligo Urban Area Small Area populations, by transit supply category.

Table 1: County Sligo: Urban Area Small Area populations by Transit Supply grouping

Urban Area	Transit Supply category							Total
	Zero Supply	Very Low	Low	Below av.	Above av.	High	Very High	
Sligo	0.0% (0)	0.0% (0)	1.8% (680)	13.7% (5,214)	17.8% (6,758)	16.5% (6,262)	4.5% (1,694)	54.3% (20,608)
Tubbercurry	0.7% (250)	0.0% (0)	2.7% (1,038)	2.7% (1,019)	0.0% (0)	0.0% (0)	0.0% (0)	6.1% (2,307)
Strandhill	0.0% (0)	0.0% (0)	0.0% (0)	3.9% (1,464)	1.4% (518)	0.0% (0)	0.0% (0)	5.2% (1,982)
Collooney	0.0% (0)	0.0% (0)	1.0% (386)	3.2% (1,209)	0.5% (202)	0.0% (0)	0.0% (0)	4.7% (1,797)
Ballysadare	0.0% (0)	0.0% (0)	0.0% (0)	3.2% (1,208)	1.4% (539)	0.0% (0)	0.0% (0)	4.6% (1,747)
Ballymote	0.0% (0)	0.7% (263)	0.0% (0)	3.8% (1,448)	0.0% (0)	0.0% (0)	0.0% (0)	4.5% (1,711)
Enniscrone	1.9% (710)	0.0% (0)	0.9% (345)	0.6% (236)	0.0% (0)	0.0% (0)	0.0% (0)	3.4% (1,291)
Coolaney	1.4% (531)	0.0% (0)	0.0% (0)	1.6% (624)	0.0% (0)	0.0% (0)	0.0% (0)	3.0% (1,155)
Rosses Point	0.0% (0)	0.0% (0)	0.0% (0)	1.0% (388)	1.3% (495)	0.0% (0)	0.0% (0)	2.3% (883)
Grange	0.0% (0)	0.0% (0)	0.0% (0)	1.5% (569)	0.0% (0)	0.0% (0)	0.0% (0)	1.5% (569)
Cliffony	0.0% (0)	0.0% (0)	0.3% (97)	1.1% (424)	0.0% (0)	0.0% (0)	0.0% (0)	1.4% (521)
Ballincar	0.0% (0)	0.0% (0)	0.0% (0)	1.1% (423)	0.0% (0)	0.0% (0)	0.0% (0)	1.1% (423)
Gurteen	0.4% (155)	0.0% (0)	0.0% (0)	0.7% (266)	0.0% (0)	0.0% (0)	0.0% (0)	1.1% (421)
Riverstown	0.0% (0)	0.0% (0)	0.5% (197)	0.5% (197)	0.0% (0)	0.0% (0)	0.0% (0)	1.0% (394)
Carney	0.0% (0)	0.0% (0)	0.0% (0)	1.0% (373)	0.0% (0)	0.0% (0)	0.0% (0)	1.0% (373)
Ballintogher	0.0% (0)	0.0% (0)	0.0% (0)	0.9% (351)	0.0% (0)	0.0% (0)	0.0% (0)	0.9% (351)

Dromore	0.0% (0)	0.0% (0)	0.0% (0)	0.8% (292)	0.0% (0)	0.0% (0)	0.0% (0)	0.8% (292)
West								
Ballygawley	0.0% (0)	0.0% (0)	0.0% (0)	0.8% (285)	0.0% (0)	0.0% (0)	0.0% (0)	0.8% (285)
Easky	0.0% (0)	0.0% (0)	0.4% (154)	0.2% (81)	0.0% (0)	0.0% (0)	0.0% (0)	0.6% (235)
Ballynacarrow	0.0% (0)	0.0% (0)	0.0% (0)	0.6% (220)	0.0% (0)	0.0% (0)	0.0% (0)	0.6% (220)
Charlestown-Bellahy	0.0% (0)	0.0% (0)	0.0% (0)	0.5% (196)	0.0% (0)	0.0% (0)	0.0% (0)	0.5% (196)
Mullaghmore	0.0% (0)	0.0% (0)	0.0% (0)	0.5% (172)	0.0% (0)	0.0% (0)	0.0% (0)	0.5% (172)
Total	4.3% (1,646)	0.7% (263)	7.6% (2,897)	43.9% (16,659)	22.4% (8,512)	16.5% (6,262)	4.5% (1,694)	100.0% (37,933)

Table 3: County Sligo: SI score & transit supply category by Urban Area, weighted by population.

There is a statistically significant variation between Counties in the share of Small Areas in each transit supply category ($\chi^2(132) = 402.61, p < .001$). Table 3 indicates that Mayo and Roscommon have higher shares of their population living in Small Areas with no transit supply (over two people in every five), whereas Sligo has the lowest (close to one in eight). Close to one-third of people living in Sligo live in Small Areas with SI scores above the average (not including areas with no transit), followed by a little under one in four in Clare and around one in seven in Roscommon. At the other end of the spectrum, only around one in twelve people in Mayo and Leitrim live in areas with above the average SI scores, while for Donegal it is only one in twenty.

Urban areas

Approximately 50% of County Sligo's population (close to 38,000 people) live in Small Areas that are within one of the 22 named 'urban areas' within County Sligo. Figure 4 shows the total population and aggregate SI scores for each urban area.

Boxplots of the SI scores for Small Areas within each of the various urban areas are shown in Figure 10, again grouped by County.

Table 1 shows summary statistics for SI scores and the population, overall for all of the County Sligo and split by transit supply category. The total population of the County Sligo is 70,198, of whom 77% (53,730) live in Small Areas where the SI score is below the average (of those Small Areas with a non-zero SI score) (377.5). This includes the 9,461 people (13%) who live in Small Areas with no transit at all.

Figure ?? shows transit Supply Index (SI) categories by Small Areas across the County Sligo. It also shows to the population living in Small Areas in each category for both the entire County Sligo and for each county. In general, there appear to be large areas of the County Sligo with SI scores below the average, or with no transit supply at all. Also, as might be expected, Small Areas where SI scores are higher than the average (of those with non-zero scores) appears to be located in very small and scattered geographic areas (at population centers).

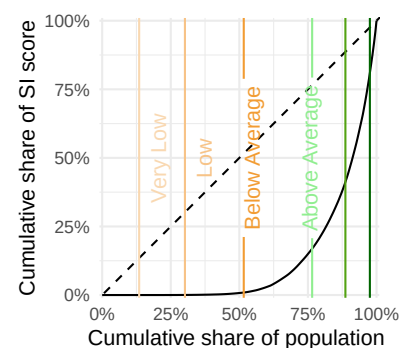


Figure 2: County Sligo: Lorenz curves

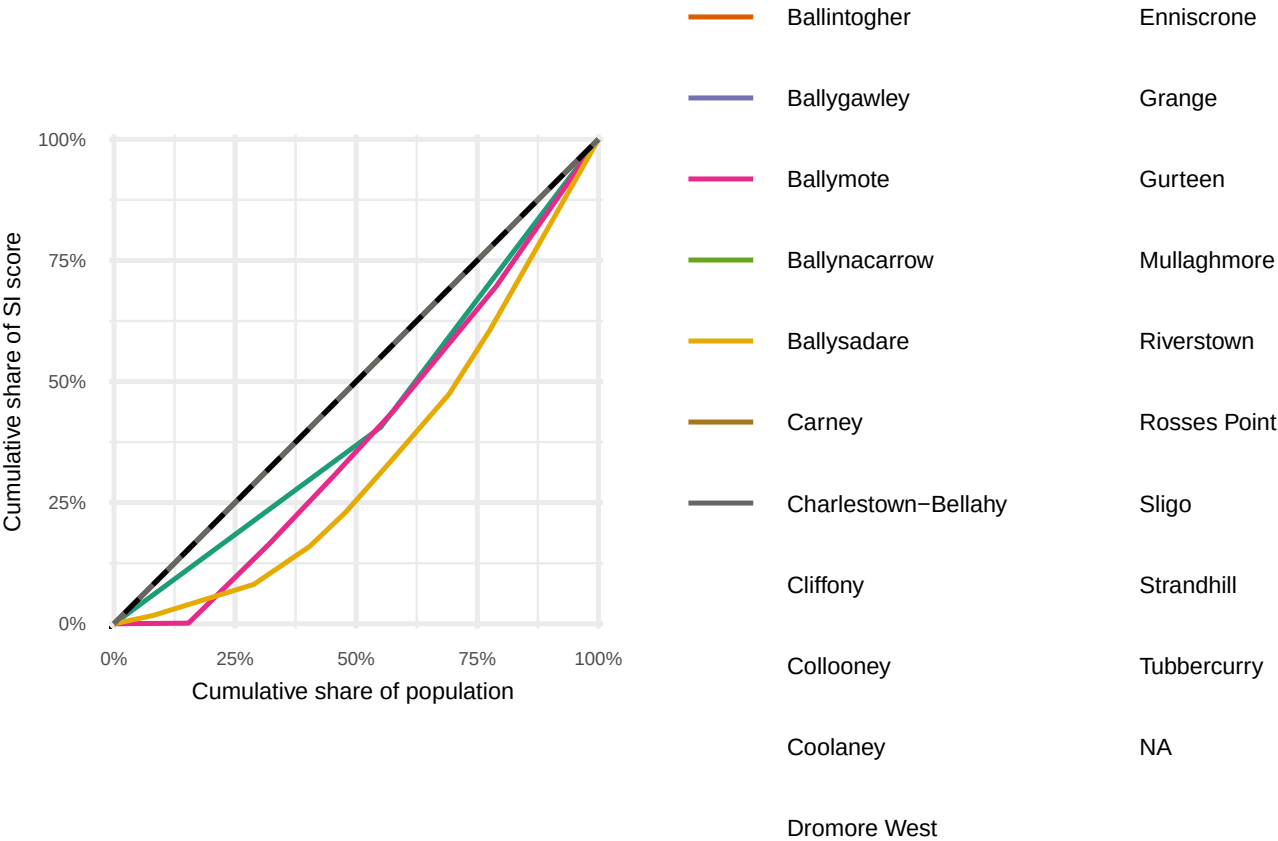


Figure 3: County Sligo: Lorenz curves by Urban Areas

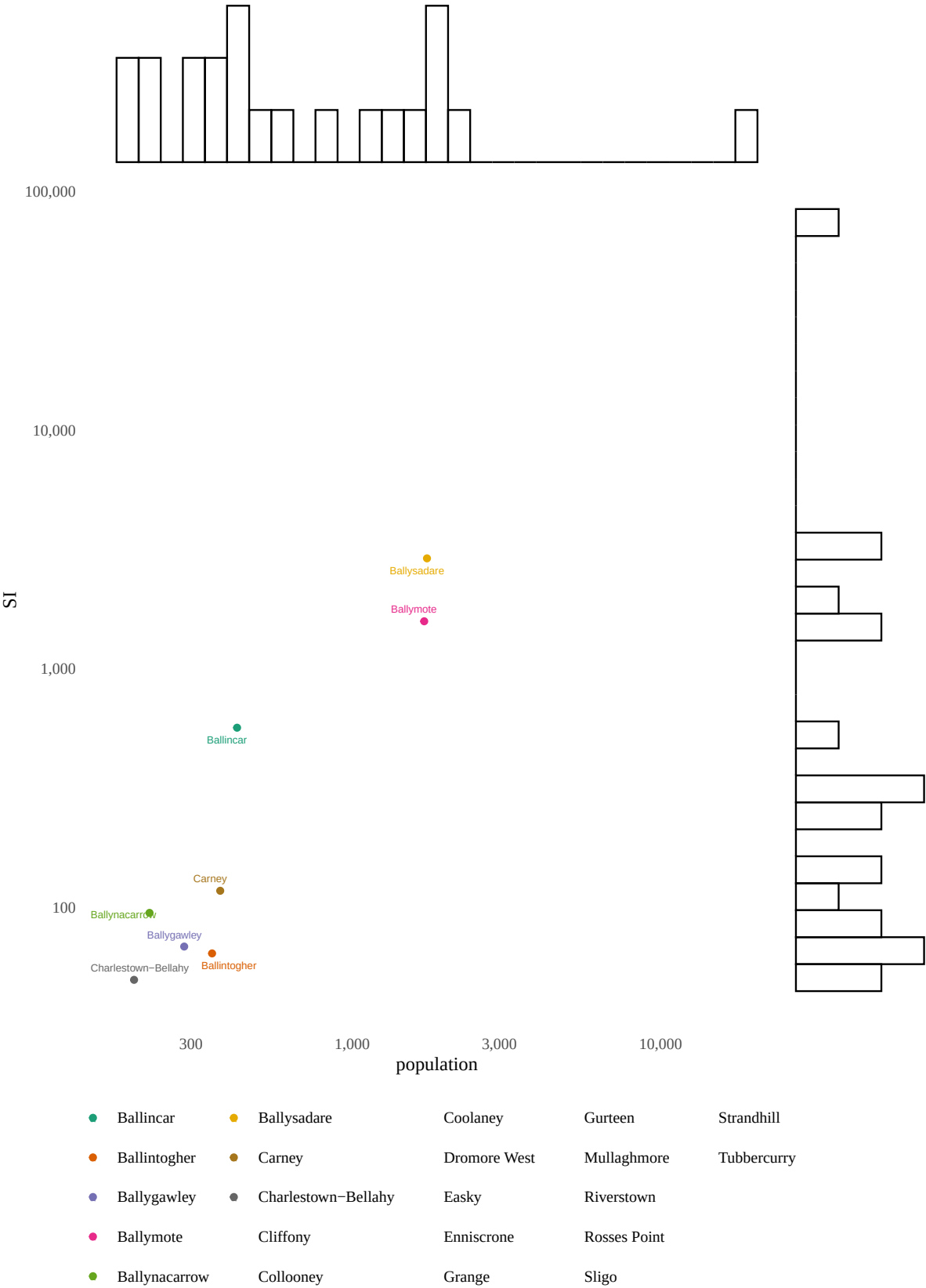


Figure 4: County Sligo pop. vs SI, aggregated by urban areas

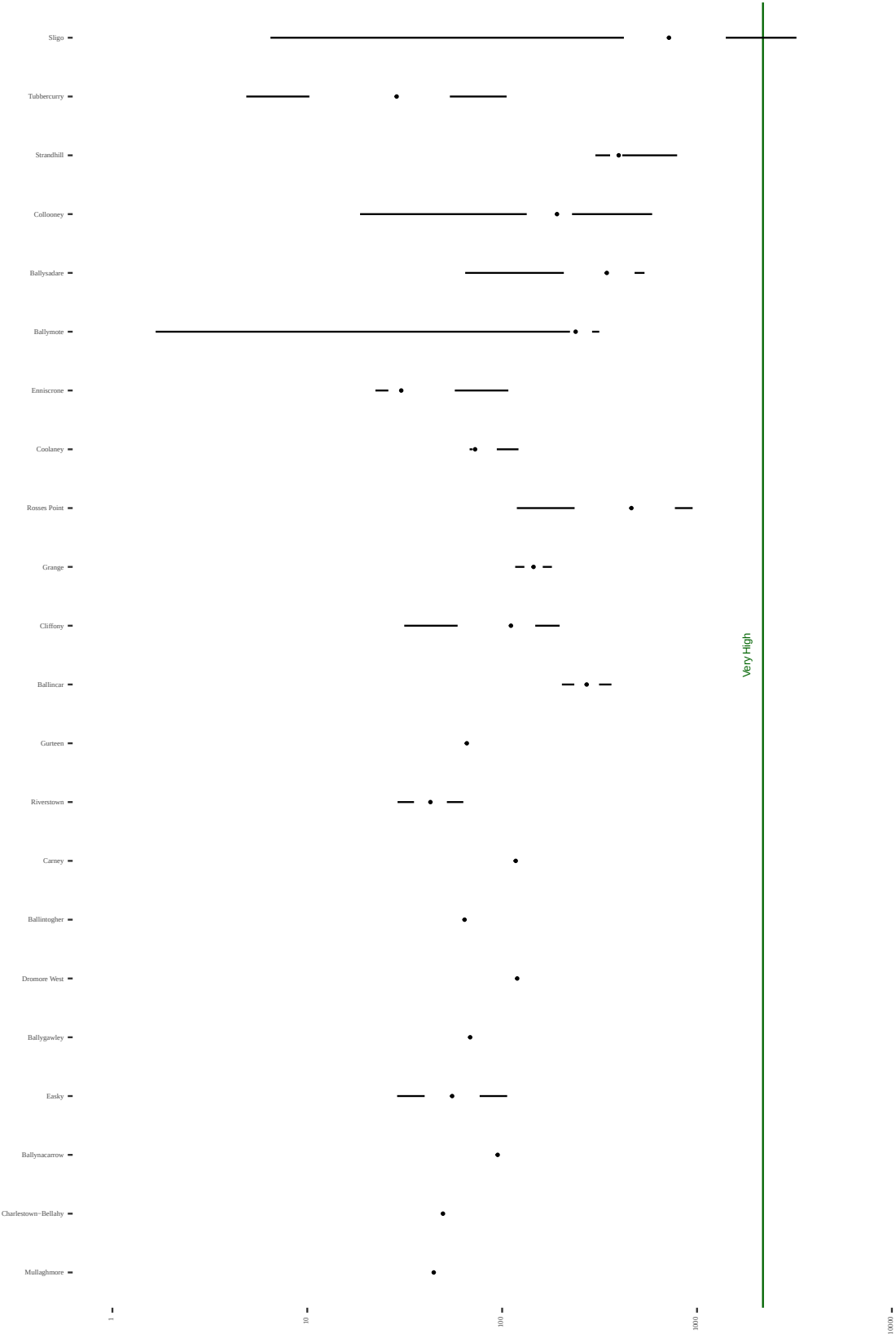


Figure 5: County Sligo urban areas: boxplot of Small Area SI scores, ordered by population size

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Dromore West	0.0% (0)	0.0% (0)	0.0% (0)	0.8% (292)	0.0% (0)	0.0% (0)	0.0% (0)	0.8% (292)
Ballygawley	0.0% (0)	0.0% (0)	0.0% (0)	0.8% (285)	0.0% (0)	0.0% (0)	0.0% (0)	0.8% (285)
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Total	4.3% (1,646)	0.7% (263)	7.6% (2,897)	43.9% (16,659)	22.4% (8,512)	16.5% (6,262)	4.5% (1,694)	100.0% (37,933)

As shown in Table 2, the largest shares of the total population of the County Sligo are the

There is a statistically significant variation between Counties in the share of Small Areas in each transit supply category ($\chi^2(132) = 402.61$, $p < .001$). Table 3 indicates that Mayo and Roscommon have higher shares of their population living in Small Areas with no transit supply (over two people in every five), whereas Sligo has

Table 3: SI score & transit supply category by County, weighted by population.

Characteristic	Overall N = 70,198	Sligo N = 20,608	Tubbercurry N = 2,307	Strandhill N = 1,982	Collooney N = 1,797	Ballysadare N = 1,747	Ballymote N = 0	Enniscrone N = 0	Coolagh N = 0
SI									
Min	0.0	6.5	0.0	300.9	18.7	64.6	NA	NA	NA
Q1	0.9	393.8	4.9	363.0	100.1	94.2	NA	NA	NA
Mean	323.0	931.5	31.3	422.7	192.0	305.2	NA	NA	NA
Q3	396.4	1,269.7	51.7	419.0	216.1	478.3	NA	NA	NA
Max	3,237.0	3,237.0	105.4	790.6	588.6	537.6	NA	NA	NA
Sum	22,671,969.6	19,196,567.0	72,231.3	837,776.9	345,000.5	533,261.2	NA	NA	NA
transit_supply									
Zero Supply	9,461 (13%)	0 (0%)	250 (11%)	0 (0%)	0 (0%)	0 (0%)	0 (NA%)	0 (NA%)	0 (NA%)
Very Low	11,714 (17%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (NA%)	0 (NA%)	0 (NA%)
Low	15,053 (21%)	680 (3.3%)	1,038 (45%)	0 (0%)	386 (21%)	0 (0%)	0 (NA%)	0 (NA%)	0 (NA%)
Below average	17,502 (25%)	5,214 (25%)	1,019 (44%)	1,464 (74%)	1,209 (67%)	1,208 (69%)	0 (NA%)	0 (NA%)	0 (NA%)
Above average	8,512 (12%)	6,758 (33%)	0 (0%)	518 (26%)	202 (11%)	539 (31%)	0 (NA%)	0 (NA%)	0 (NA%)
High	6,262 (8.9%)	6,262 (30%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (NA%)	0 (NA%)	0 (NA%)
Very High	1,694 (2.4%)	1,694 (8.2%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (NA%)	0 (NA%)	0 (NA%)

¹ n (%)

the lowest (close to one in eight). Close to one-third of people living in Sligo live in Small Areas with SI scores above the average (not including areas with no transit), followed by a little under one in four in Clare and around one in seven in Roscommon. At the other end of the spectrum, only around one in twelve people in Mayo and Leitrim live in areas with above the average SI scores, while for Donegal it is only one in twenty.

A Lorenz Curve of distribution of the County Sligo's transit supply to the population is shown in Figure 2. The Gini co-efficient is 0.76, indicating a relatively unequal distribution. In total, the SI score across all of the County Sligo is approximately 100,000, but NA of this goes to Small Areas that are home to just NA of the total population. NA of the population live in Small Areas that, in combination, receive just NA of the total supply. This is perhaps not surprising, given that providing transit services in rural and low density areas is challenging.

Figure 7 shows Lorenz curves for each of the named urban areas. Gini co-efficients range from

Urban areas Approximately 50% of the County Sligo's population (close to 38,000 people) live in Small Areas that are within one of the 22 named 'urban areas' within the west. Figure 8 shows the total population and aggregate SI scores for each urban area, while Figure ?? shows the same data, but grouped by County

Boxplots of the SI scores for Small Areas within each of the various urban areas are shown in Figure 10, again grouped by County.

County Sligo

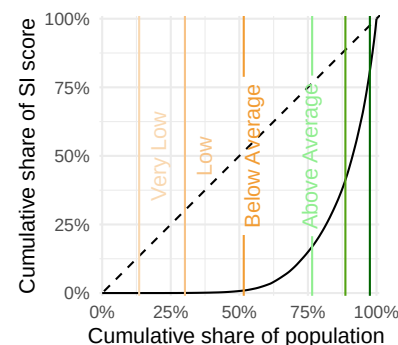


Figure 6: County Sligo: Lorenz curves

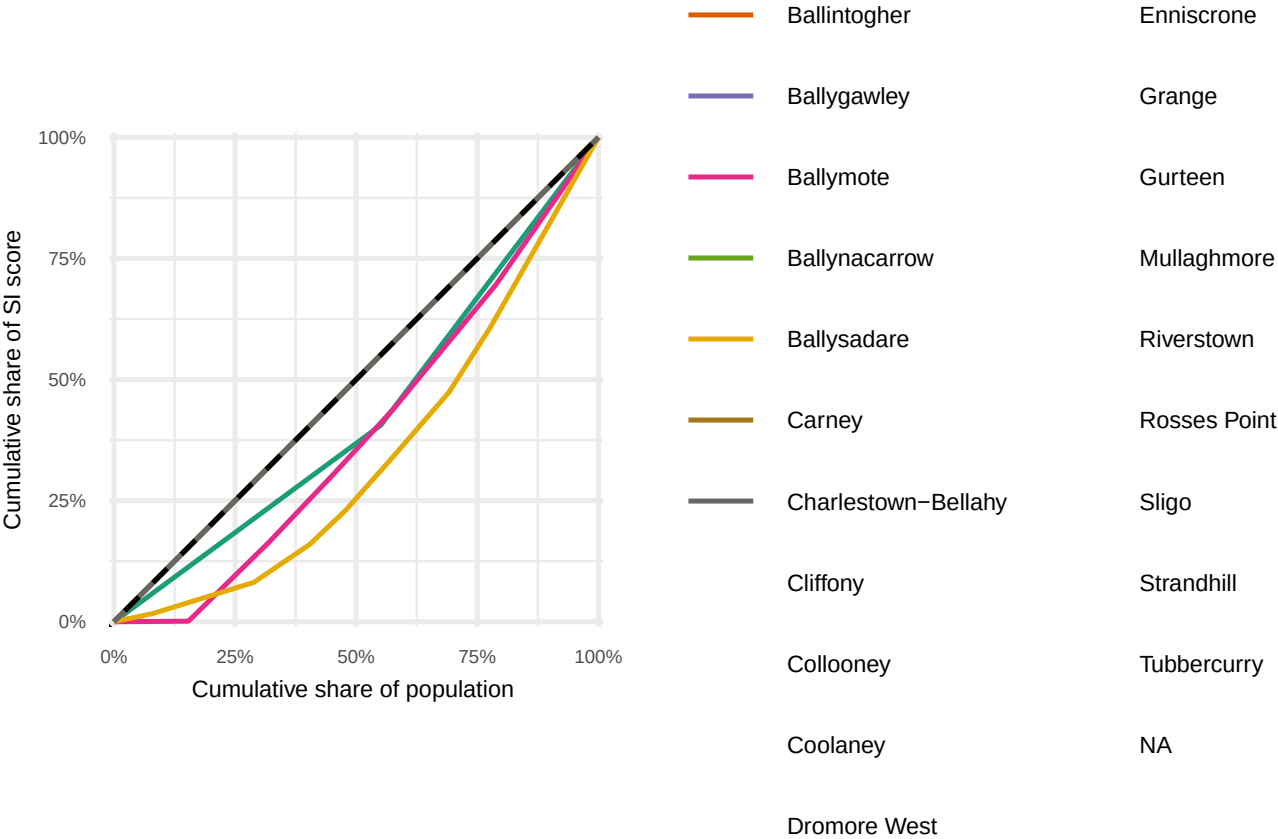


Figure 7: County Sligo: Lorenz curves by County

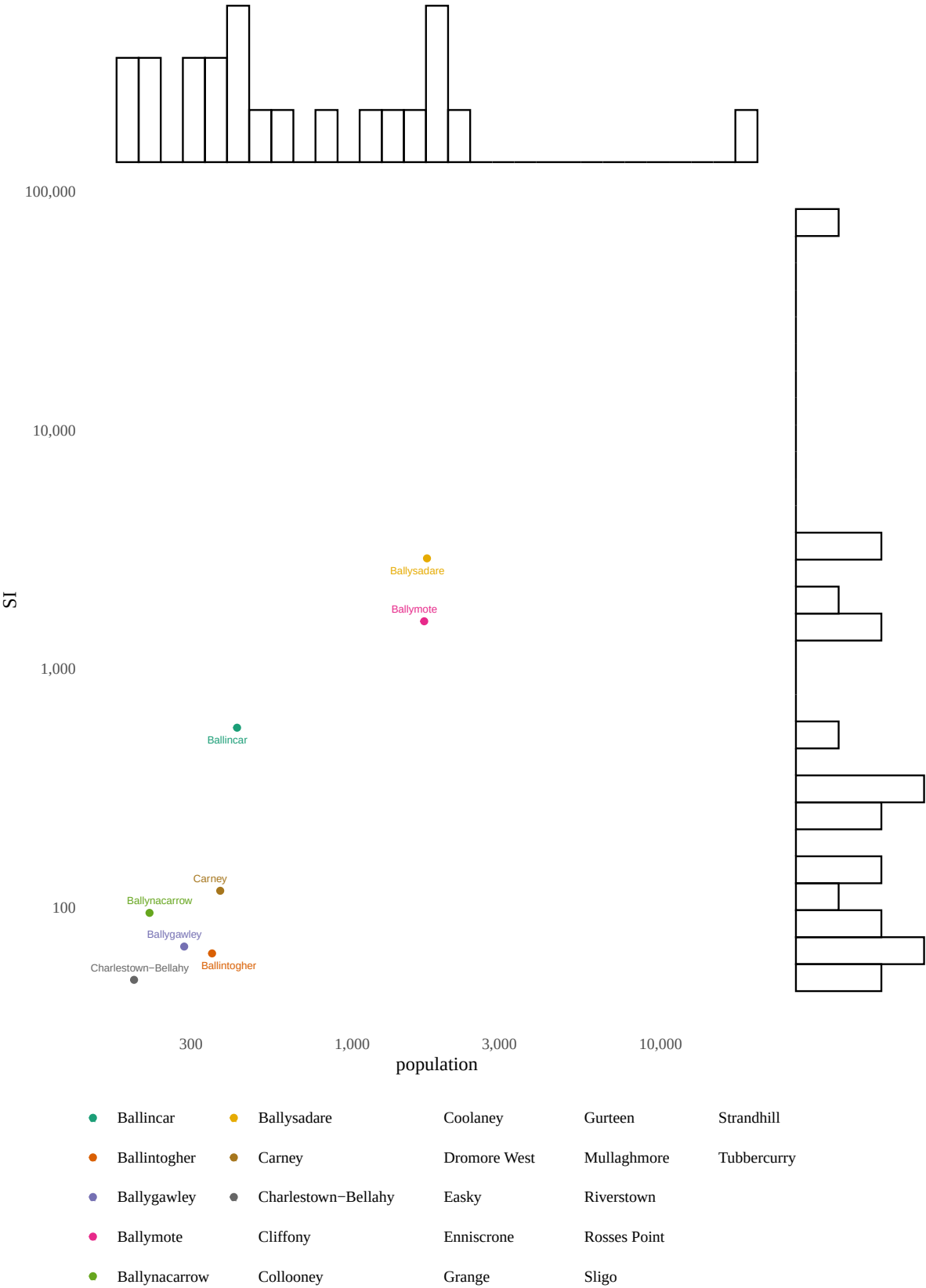


Figure 8: County Sligo pop. vs SI, aggregated by urban areas



Figure 9: County Sligo pop. vs SI, aggregated by urban areas

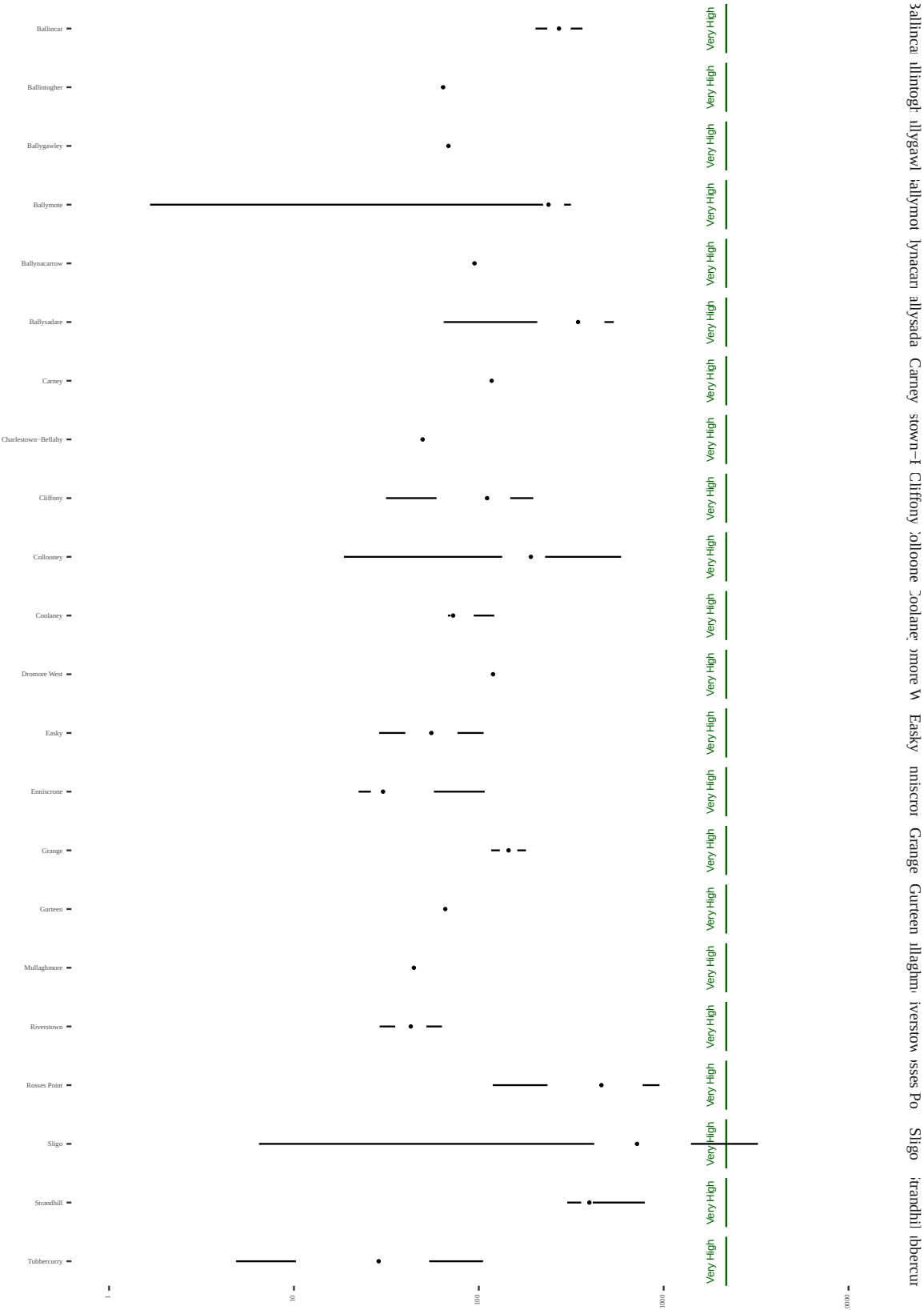


Figure 10: County Sligo urban areas: boxplot of Small Area SI scores, ordered by population size